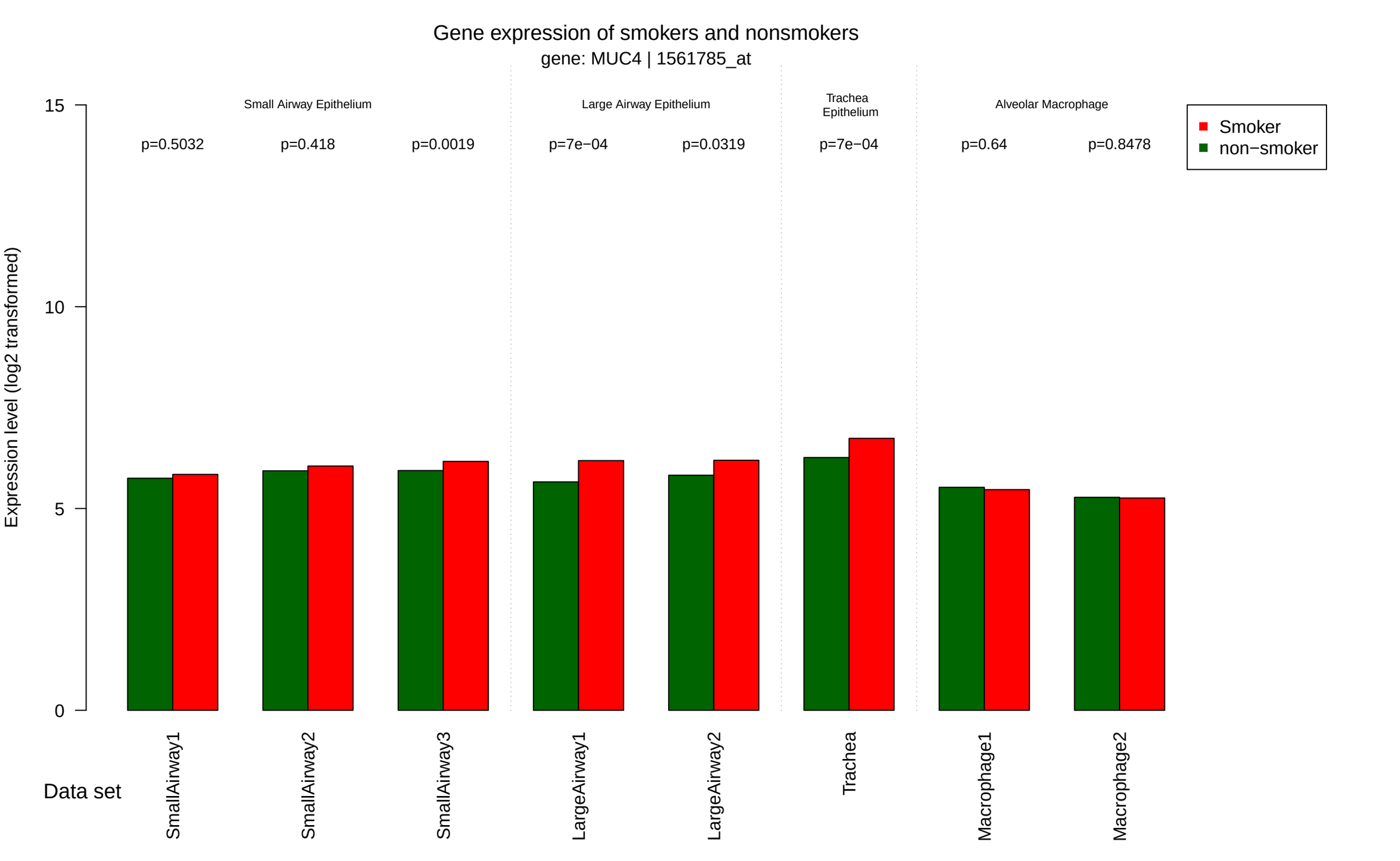
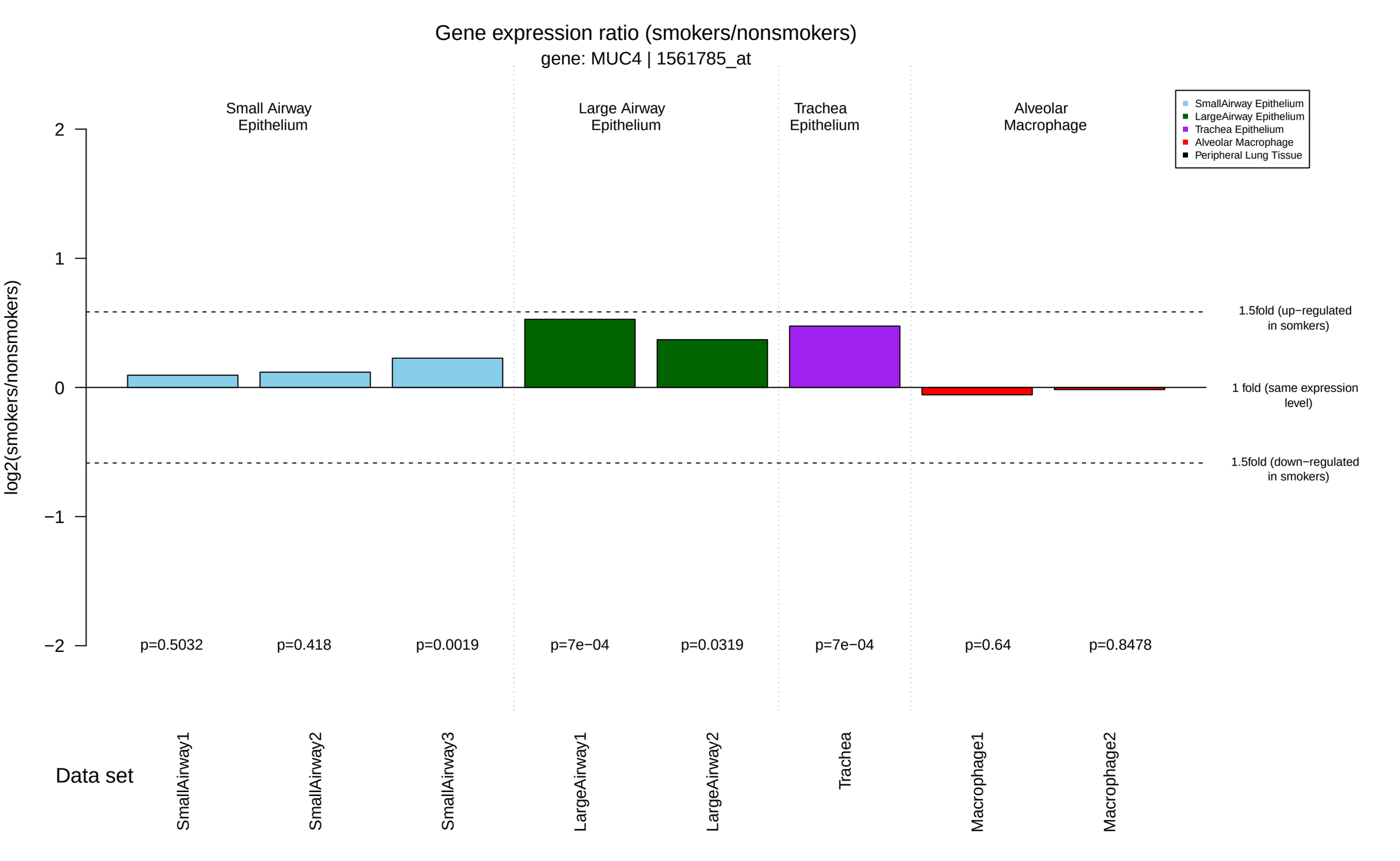
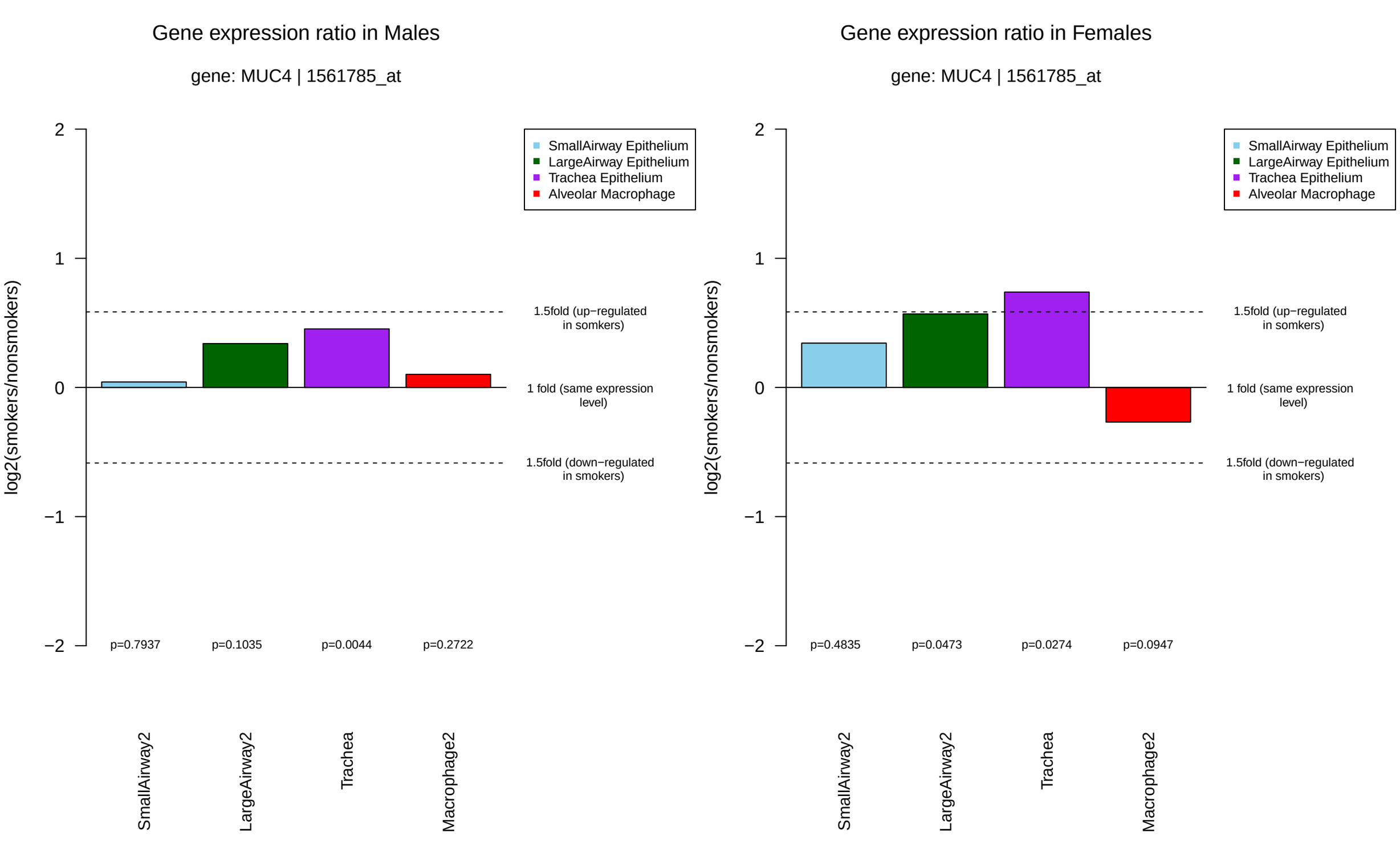
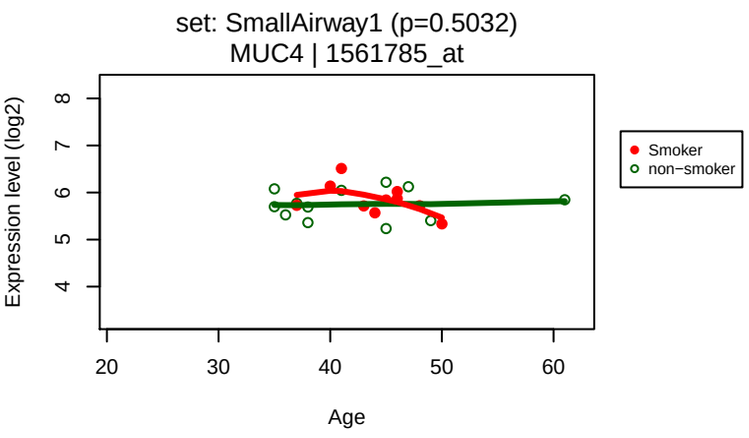
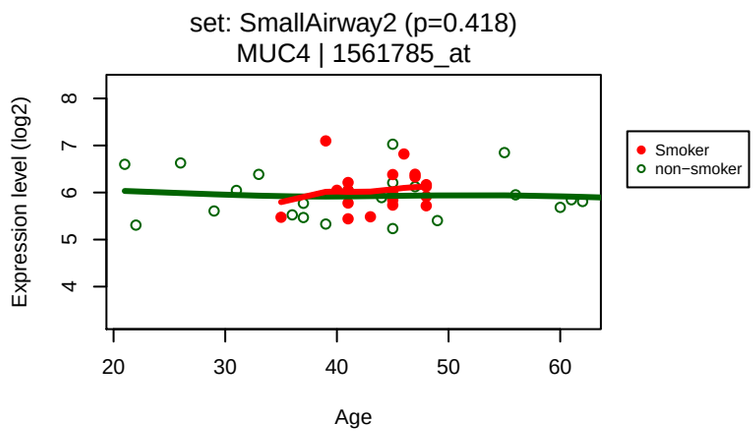
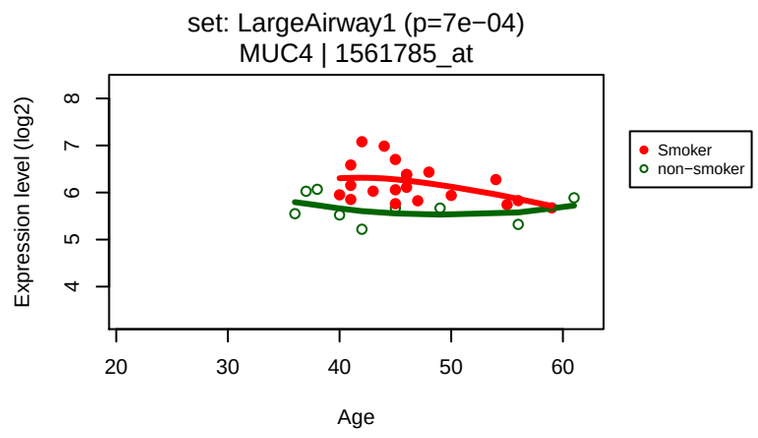

Gene expression of smokers and nonsmokers
gene: MUC4 | 1561785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5032
p=0.418
p=0.0019
p=7e−04
p=0.0319
p=7e−04
p=0.64
p=0.8478
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

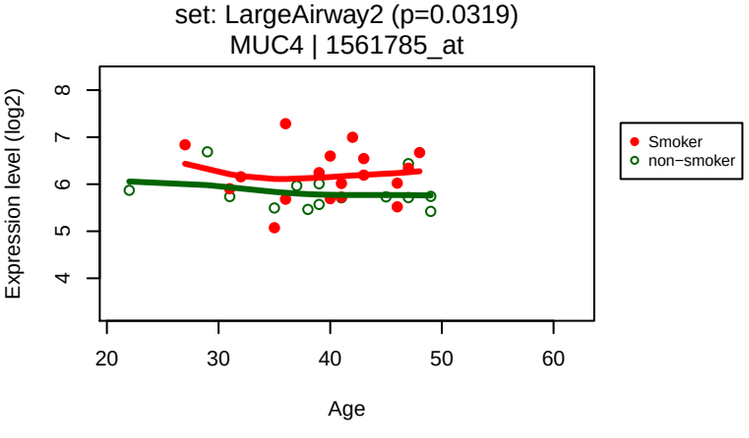

Gene expression ratio (smokers/nonsmokers)
gene: MUC4 | 1561785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5032
p=0.418
p=0.0019
p=7e−04
p=0.0319
p=7e−04
p=0.64
p=0.8478
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

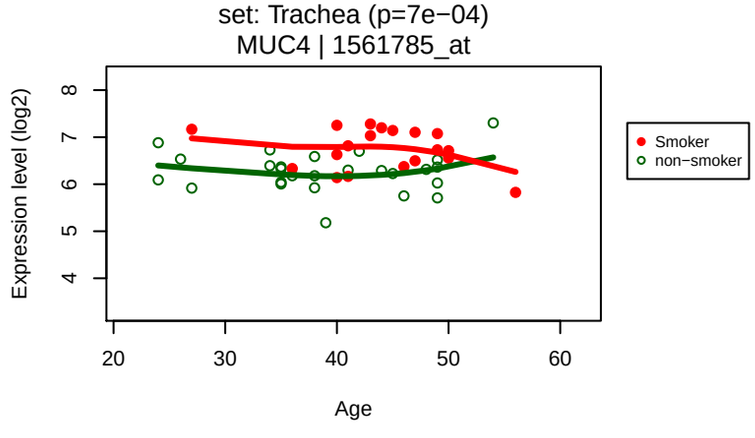

Gene expression ratio in Males
gene: MUC4 | 1561785_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7937
p=0.1035
p=0.0044
p=0.2722
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC4 | 1561785_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4835
p=0.0473
p=0.0274
p=0.0947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

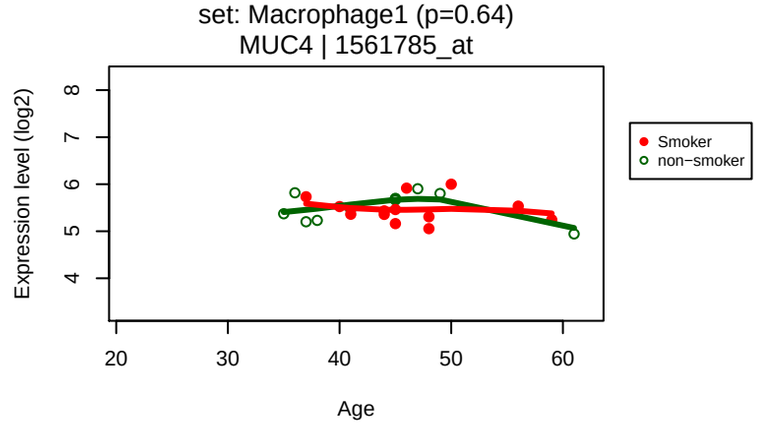

set: SmallAirway1 (p=0.5032)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker

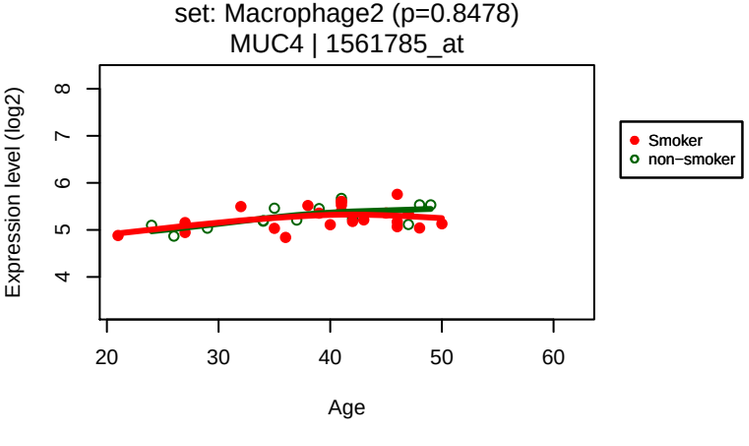

set: SmallAirway2 (p=0.418)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker

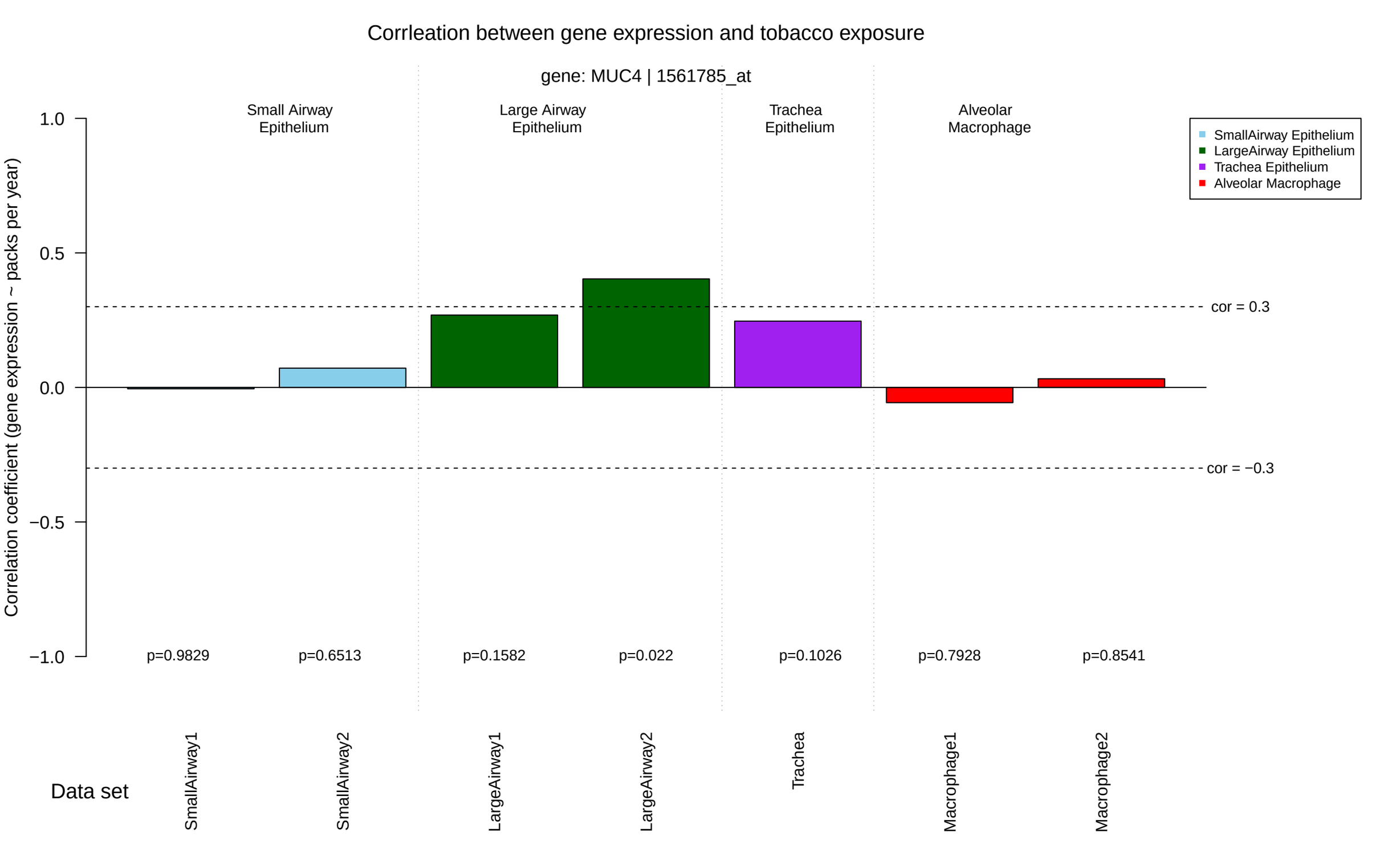

set: LargeAirway1 (p=7e−04)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0319)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=7e−04)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.64)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8478)
MUC4 | 1561785_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC4 | 1561785_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9829
p=0.6513
p=0.1582
p=0.022
p=0.1026
p=0.7928
p=0.8541
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2